
A Phase-Space Method for Arbitrary BiMolecular Gas-Phase Reactions: Theoretical Description

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ABSTRACT: A theoretical model for the calculation of rate constants for arbitrary bimolecular gas-phase reactions was developed. The method is based on the phase-space statistical method developed by Light and co-workers [1–6]. In the present article this method is extended to arbitrary molecular systems. The new method requires knowledge of the molecular properties in the reaction and products channels of the chemical system. The properties are the vibrational frequencies, moments of inertia, and potential energy for the interacting species in their ground state equilibrium configuration. Furthermore, we have to calculate either the energy barrier or the long-range potential for the chemical system (if the reaction channel does not have an energy barrier). The usefulness of the method is that it can be applied to all bimolecular reactions, trimolecular reactions, and even reactions of higher orders. Therefore, it can be applied to cases where rate constants of complex chemical reactions are required, but reliable laboratory measurements or other means to estimate rate parameters are not yet possible. Even if spectroscopic data are not available for the reactants and products, it is possible to use electronic structure theory to calculate the required data. © 2001 John Wiley & Sons, Inc. *Int J Quantum Chem* 84: 479–492, 2001

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Introduction

Within the applied scientific areas of atmospheric chemistry, combustion, and pyrolysis (hydrocarbon cracking), elementary chemical reactions are fundamental components of theoretical models. There is a critical need for improving the available mechanistic data for research in these areas. Currently, many mechanistic parameters are estimated by analogy with known reactions or through empirical structure–reactivity relationships. Of particular importance are the reactions of HO_x with alkanes, alkenes, aromatics, and aldehydes; the interactions of HO_2 , RO_2 , and NO_3 radicals with each other; and the decomposition of alkyl and alkoxy radicals. Many of the chemical reactions involved in these models can be characterized as large molecule systems (LMS).¹

Over the years, reaction kinetics has been an experimentally based science due to large limitations related to theoretical models, but there are many examples from the literature which indicate that the reactivity of organic compounds still gives rise to a major source of uncertainties in the chemical mechanism. One such area is the elementary bimolecular chemical reactions of atmospheric chemistry models [7–14]. Important examples include the chemistry of higher molecular weight organic compounds, peroxy radical reactions, and photo-oxidation processes, i.e., quantum yields and absorption cross sections. The uncertainties in rate parameters are usually greater for the more reactive organic compounds. Most rate parameters are frequently measured at temperatures near 298 K due to experimental difficulties. The temperature dependence of many reactions is not well characterized as can be seen in the standard data bases: NIST [15], IUPAC [16], NASA [17]. A complete data base for atmospheric chemistry, combustion, and pyrolysis needs rate parameters that are valid over a temperature range of 150 to greater than 2000 K.

Although the progress in computer technology has increased remarkably over the years, the bottleneck of calculating rate constants based upon full dynamical methods is the calculation of a reliable PES for the system. Therefore, much effort has been

devoted to the development of models that need less information about the PES.

Theoretical models can be grouped into three different classes:

- dynamical,
- statistical-dynamical, and
- statistical methods.

Furthermore, dynamical methods can be grouped into

- fully dynamical and
- reaction path methods.

The benchmark in dynamical methods is the Hamilton operator. When no exterior forces act on a chemical reaction, the quantum mechanical Hamilton operator is simply equal to the sum of the kinetic energy operator and the potential energy.

Since the 1930s, the relationship between PESs and the dynamics of molecular/atomic collisions has been known—this is the numerical solution to the classical or quantum mechanical equations of motion. An analysis of the outcome of the molecular/atomic collisions can be used, e.g., to calculate the reaction probability of the reaction. From these results, the total rate constant is obtained by integrating over the impact parameter and averaging over the Boltzmann distribution.

If the motion of the system of nuclei is treated quantum mechanically, the accuracy of the calculation is only limited by the accuracy of the PES, and the number of basis function used to expand the wave function that describes the system. Unfortunately, fully quantum mechanical simulations demand enormous computational resources, even for triatomic systems. Since a complete quantum mechanical treatment of most molecular systems is computationally expensive—or impossible—mixed quantum-classical or purely classical treatments are very popular for solving various inelastic, elastic, and reactive scattering problems.

If a PES is available, classical trajectory methods can always be used since only the coordinates and momenta of the nuclei describe the dynamics. Such calculations are not very time consuming. The coordinates and momenta of the nuclei are functions of time; therefore, in classical dynamics, one follows the system as it goes from reactants to products during a collision. The outcome of a single trajectory in classical dynamics is not very meaningful. In-

¹The term LMS will, in this article, define systems where it is not possible to evaluate reliable potential energy surfaces (PES) or estimate the transition state configurations using electronic structure theory.

stead, one has to average over a large number of trajectories to obtain reliable results. Classical trajectories are valid when the momentum is large (i.e., the relative speed), the energy is high, the colliding nuclei are heavy, and the temperature of the system is not too low. In general, classical trajectory computations are often accurate for calculating total rate constants if a reliable PES of the chemical system is known.

In reaction path methods, the dynamics of the molecular motion are studied along the reaction path (the minimum-energy part on the PES). As indicated by the name of the model, in contrast to the fully dynamic methods, only PES information along the reaction path is required. Therefore, the Hamilton operator is only required in this region. Both classical and quantum-classical theories of these methods have been developed; for further details, see Ref. [18].

Regarding dynamical methods the necessity of potential energy surface limits the applicability of dynamic methods and justifies the use of more approximative methods such as transition state theory (TST), variation transition state theory (VTST), and phase-space theory (PST).

Applying TST to a chemical reaction gives the total rate constant for that reaction. The basic assumption of TST is that a chemical reaction takes place by forming an activated complex (transition state) which is in equilibrium with the reactants. In TST, the activated complex is positioned at the top of the activation barrier, and it is assumed that once the system has crossed the transition state configuration, it does not return. In order to use TST, one only needs detailed spectroscopic and potential energy information about the reactants and the transition state. Even though TST needs much less PES information, it is only possible to obtain the necessary PES information for a very limited number of molecular systems.

In VTST, the usual transition state expression is evaluated along the reaction path, and the VTST rate constant is obtained where the transition state rate has its minimum. This corresponds to the situation where the flux from reactants to products is a minimum. This method requires PES information for the whole reaction path, and is therefore less applicable than TST.

PST requires information about the reactant and product channels of the chemical reaction, vibrational frequencies, moments of inertia, potential energy in their ground state equilibrium configuration, and the energy barrier or the long-range

potential (if the chemical reaction does not have an energy barrier). But this can be calculated with sufficient accuracy using electronic structure theory. Moreover, state (rot-vib) resolved and total cross sections and rates can be calculated within this approach. PST can be used for all known chemical reactions as long as all the species in the reactant and product channels are known.

All these models except PST require that the transition state of the system is known. Finding the reaction path for a system after the transition state configuration is localized is a straightforward task in many cases. On the other hand, locating the first order saddle point (the transition state configuration) is much more difficult. There exists no general method which can be used to find the transition state, and there is no guarantee that only one transition state between the reactants and products exists. Furthermore, the reaction path tends to become more complicated as the number of nuclei in the chemical system increases (e.g., LMS) and the possible number of transition states increase.

Therefore, we have developed a theoretical method which can be used to calculate rate constants for reactions involving larger organic species, e.g., systems with high molecular weight. The method is based on the phase-space statistical methods described by Light who has developed a model based upon phase-space (statistical) theory to calculate rate constants for different types of atom-diatom molecular exchange reactions [1–6]. The theory has been applied to several different interactions of atoms with diatomic molecules: ion-molecule [1–3], atom-excited molecule [4], atom-molecule reactions with no activation energy [4], and atom-molecule reactions with activation energy [5]. The disadvantage of the statistical approach is that it does not include any dynamical effects. Therefore it is expected that the method will only be able to estimate the order of magnitude of the rate constants. However, it has been shown by Lin and Light [5] that phase-space methods are able to predict the temperature dependence rather accurately.

Phase-space methods have been applied to ionic fragmentation [19–21], where the following dissociation products are considered: sphere-atom, two linear, linear-atom, two spheres, or sphere-linear. Klippenstein and Marcus [22] have applied phase-space methods to the unimolecular dissociation process of CH_2CO . Finally, Chesnavich and Bowers [23, 24] have derived the classical formulas for the sum and

density of rotational and rotational-orbital states for polyatomic pair molecules, for the pair combinations: sphere-linear, sphere-sphere, and sphere-symmetric. Chesnavich and Bowers examined the $\text{C}_2\text{H}_4^+ + \text{C}_2\text{H}_4 \rightarrow \text{C}_3\text{H}_5^+ + \text{CH}_3$ reaction under the assumption that the C_3H_5^+ molecule can be considered as a symmetric top molecule. However, rate constants were not calculated; in their paper, they only examined the reactant's and product's phase-spaces and the process cross section and product kinetic energy distribution.

In this article, we have extended the method by Light to arbitrary molecular systems. We have replaced the sums in the evaluated equations by integrals to a much lower extent than done in the previous applications of the phase-space technique. The new method requires the vibrational frequencies and moments of inertia and potential energy for the reacting and product species in their ground state equilibrium configuration. Moreover, the size of the energy barrier, or the long-range potential for the chemical systems if the reaction channel has no energy barrier, must be determined. That means that reaction path, transition state information, etc., are not required.

The next section contains our theoretical description of phase-space theory to derive expressions for the cross sections and rate constants for arbitrary molecular systems. Application of this method requires calculation of the cross section for the formation of the collision complex (CC) and its subsequent decomposition through its reaction channels. Then we describe the requirements for assigning a maximum orbital angular momentum for the model.

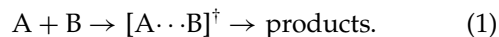
The total number of accessible states of the CC as it forms and decomposes is an important term in the equations presented, therefore, the calculation procedure for this term is given for different kinds of molecular systems. A closed expression for the rotational energy of molecules with asymmetric top symmetry does not exist, therefore, the term for this case cannot be handled in a simple manner. Fortunately, experiments and theoretical calculations show (see, e.g., Ref. [25]) that the size of two of the principal momenta of inertia are almost the same for most asymmetric top molecules. Therefore, we have treated asymmetric top molecules as slightly asymmetric top molecules, and we apply the rotational energy power series given by Polo [26].

In a forthcoming paper by Gross et al. [25], we show that the phase-space method introduced here

is superior to the correlation methods described by Wayne et al. [27].

Phase-Space Theory

We consider the problem of a molecule A composed of N atoms colliding with a molecule B composed of M atoms as



We assume that the process described by Eq. (1) takes place through different channels, α, β, γ , etc., which means that α describes the channel from the reactants to form the complex $[\text{A} \cdots \text{B}]^\ddagger$; and β, γ , etc., are accessible channels to form products from the complex. The initial and final states (or channels) of such a system are specified by the molecules' vibrational quantum numbers, $\bar{v}_i$ and $\bar{v}_f$,² rotational quantum numbers, $(\bar{j}^i, \bar{j}_z^i)$ and $(\bar{j}^f, \bar{j}_z^f)$, the relative translation energy $(E_{\text{trans}}^i, E_{\text{trans}}^f)$, and finally the orbital angular momentum quantum numbers (l^i, l_z^i) and (l^f, l_z^f) . The motion of the center of mass of A relative to B does not affect the statistics, therefore it is ignored.³

Phase-space methods in kinetics can be constructed in several ways. We will here give a derivation based upon quantum mechanics. In order to use phase-space methods, a CC $([\text{A} \cdots \text{B}]^\ddagger)$ is defined based upon two assumptions: formation (assumption III) and decomposition (assumption II) of CC. The following assumptions are made in the model.

Assumption I: The conservation laws and detailed balancing must be obeyed. Hence, the conservation laws are conservation of energy, total angular momentum, and its projection on one axis, as well as linear momentum.

The total energy of the reactants in Eq. (1) is

$$E_{\text{tot}} = E_{\text{trans}}^{A,B} + E_{\text{vib}}^A + E_{\text{vib}}^B + E_{\text{rot}}^A + E_{\text{rot}}^B + V. \quad (2)$$

The first term is the kinetic energy of the relative translational motion between the center of mass of the two molecules, A and B. The second and

² i relates to the quantum numbers of the reactants A and B, and f relates to the quantum numbers of the products.

³The number of quantum numbers required to describe a molecular state depend on the symmetry of the molecule. The description given in this section can be applied to every molecular system. Therefore, in order to give a general description, we have used the compact form $\bar{v}^x$, where x is either i or f , to describe the total number of rotational and vibrational quantum numbers for all the molecules in the reaction channels.

TABLE I

Expressions for the quantum mechanical vibrational energy for various molecules using the harmonic oscillator approximation. $\hbar$ is Planck's constant divided by 2π .

Diatomic Molecule (one oscillator)

$$E_{vib} = (v + \frac{1}{2})\hbar\omega, \text{ where } \omega \text{ is the vibrational frequency for the diatomic molecule and } v = 0, 1, 2, \dots$$

Linear Triatomic Molecule (BCD) (four oscillators)

$$E_{vib} = (v_1 + \frac{1}{2})\hbar\omega_1 + (v_{2a} + v_{2b} + 1)\hbar\omega_2 + (v_3 + \frac{1}{2})\hbar\omega_3 = E_1 + E_{2a} + E_{2b} + E_3, \text{ where } \omega_i \text{ are the four vibrational frequencies, } v_i = 0, 1, 2, \dots, \text{ and } i = 1, 2a, 2b, \text{ or } 3.$$

Spherical, Symmetric, and Asymmetric Top Molecule: $s = 3n - 6$ oscillators

$$E_{vib} = \sum_i^s (v_i + \frac{1}{2})\hbar\omega_i = \sum_i^s E_i, \text{ where } \omega_i \text{ is the vibrational frequencies for the } s \text{ oscillators and } v_i = 0, 1, 2, \dots$$

third terms are the vibrational energies of A and B, respectively. The fourth and fifth terms are the rotational energies of A and B, respectively. For different types of molecules, expressions for the ro-vibrational energy can be seen in Tables I and II. The last term, V , is the potential energy of the system.

The total angular momentum of a molecular system is a summation of the orbital angular momentum and the rotational angular momenta which gives for the reactants in Eq. (1)

$$\vec{J} = \vec{L}^{A,B} + \vec{J}^A + \vec{J}^B. \quad (3)$$

The first term is the orbital angular momentum, and the second and third terms are the rotational angular momenta of A and B, respectively.⁴

⁴Throughout this article, $\vec{a}$ means vector a , while $a = \|\vec{a}\|$.

Assumption II: In order to form a collision complex, two conditions must be fulfilled. First, it is required that the total energy of the reacting molecules can exceed the barrier of the effective potential energy. Hence, the effective potential energy is the sum of the intermolecular potential and centrifugal energy $[(L^i)^2/2\mu_{A,B}R^2]$, where $\mu_{A,B}$ is the reduced mass of the A-B system, R is the distance between the center of mass of the molecules, and L^i is the orbital angular momentum given by $(L^i)^2 = l^i(l^i + 1)\hbar^2$ ($\hbar$ is Planck's constant divided by 2π). Second, if the kinetic energy of the A-B system is sufficient to cross the barrier, there will exist a maximum value of L^i , L_{max}^i , for which scattering occurs. Thus, the probability of formation of a CC from every channel is one, if the reacting molecules can exceed the barrier of the effective potential energy and L^i is lower than or equal to L_{max}^i , otherwise the probability is

TABLE II

Expressions for the quantum mechanical rotational energy for various molecules using the rigid rotor approximation. $\hbar$ is Planck's constant divided by 2π . For the polyatomic molecules a , b , and c describe the three mutually perpendicular principal axes.

Diatomic, Triatomic, and Spherical Top Molecule

$$E_{rot} = Cj(j+1), \text{ where } C = (\hbar^2/2I)j(j+1), I \text{ is the moment of inertia for a diatomic, triatomic, or spherical top molecule, and } j = 0, 1, 2, \dots$$

Symmetric Top Molecule

$$E_{rot} = C_a j(j+1) + (C_c - C_a)k^2 = E_{rotj} + E_{rotk}, \text{ where } C_a = \hbar^2/2I_a \text{ and } C_c = \hbar^2/2I_c. \text{ If the symmetric top is prolate, the principal moments of inertia are } I_a < I_b = I_c; \text{ and if the top is an oblate, the principal moments of inertia are } I_a = I_b < I_c, j = 0, 1, 2, \dots, \text{ and } k = 0, \pm 1, \pm 2, \dots, \pm j$$

Asymmetric Top Molecule [26]

$$E_{rot} = C_1 j(j+1) + C_2 k^2 + C_3 (k^2 \sum_{n=2}^{\infty} \varepsilon_{0n} C_3^n + j(j+1) \sum_{n=1}^{\infty} \varepsilon_{1n} C_3^n + j^2(j+1)^2 \sum_{n=2}^{\infty} \varepsilon_{2n} C_3^n + \dots) = E_{rotj}^1 + E_{rotk}^1 + E_{rotj}^2 + E_{rotj}^3 + E_{rotj}^4 + \dots, \text{ where } I_c > I_b > I_a, a = \hbar^2/2I_a, b = \hbar^2/2I_b, \text{ and } c = \hbar^2/2I_c. \text{ The } \varepsilon \text{ constants are given in Ref. [26]. For the near-prolate case } (b \approx c) C_1 = (b+c)/2, C_2 = a - (b+c)/2, \text{ and } C_3 = (b-c)/2(2a-b-c). \text{ For the near-oblate case } (a \approx b) C_1 = (a+b)/2, C_2 = c - (a+b)/2, \text{ and } C_3 = (a-b)/2(a+b-2c). k_c = j \text{ when } k = 0; k_c = j - k \text{ when lower; and } k_c = j - k + 1 \text{ when upper.}$$

zero. In statistical theory, the CC is specified by J and J_z , and E_{tot} . Therefore, the probability of forming a complex at a specific (J, J_z, E_{tot}) from the initial state $(\bar{v}^j, l^i, E_{trans}^i)$ will be

$$\frac{1}{(2l^i + 1) \prod_k^{A,B} (2j^k + 1)} \quad \text{if } \bar{J} \text{ and } E_{tot} \text{ is conserved} \\ 0 \quad \text{otherwise} \quad (4)$$

which gives

$$\sigma(\bar{v}r^i, l^i; J, J_z, E_{tot}) = \frac{\pi \hbar^2}{2\mu_{A,B} E_{trans}^i} \\ \times \sum_{\left\{ \begin{array}{l} l^i \\ \bar{J} \text{ and } E_{tot} \text{ conserved} \\ l^i \leq l_{max}^i \end{array} \right\}} (2l^i + 1) \frac{(2J + 1)}{(2l^i + 1) \prod_k^{A,B} (2j^k + 1)}. \quad (5)$$

In Eq. (5), the summation over l^i is restricted to three constraints: conservation of total energy (Assumption I); conservation of total angular momentum (Assumption I); and the constraint $l^i \leq l_{max}^i$ (Assumption II). Although for atom-diatomic systems the "triangular" inequality $|l_{atom-diatom} - j_{diatom}| \leq J \leq |l_{atom-diatom} + j_{diatom}|$ will ensure conservation of angular momentum, the conservation of angular momentum cannot be treated in such a simple manner because we are dealing with systems composed of a larger number of atoms; this problem will be discussed later [cf. "Evaluation of $N(E_{tot}, J)$ "].

Assumption III: Decomposition of CC is uncorrelated with the modes of the reactants except through conservation laws and detailed balancing. That means the decomposition of CC into the different channels is of equal probability.

Assumption I is required because energy and momentum must be conserved, and the condition of detailed balance must be fulfilled to have a meaningful model of a chemical system. On the other hand, Assumptions II and III are rather rough approximations—classical, quantum-classical, and quantum mechanical dynamical simulations show clearly that the formation and decomposition of a CC cannot be considered to be that simple [18]. A CC can also be formed when the system's kinetic energy is lower than the effective potential energy. One example is that a CC is formed due to resonance behavior of the collision between A and B, e.g., due to tunneling through the barrier [28]. Another alternative under these conditions is the formation of nonresonance complexes. A CC formed under these conditions has a very small probability of tunneling through the barrier [28]. These two

cases will not contribute essentially to the reaction probability around room temperature, if the atoms taking part in the collision are heavy particles. Assumption II is a reasonable approximation for most of the LMS that we wish to investigate with phase-space methods. Assumption III is reasonable for reactions where the lifetime of a CC is long, then there is an equal probability of the CC decomposing into the available manifolds of the different reaction channels. On the other hand, if the reaction is very fast, our method will overestimate the rate constant.

These three assumptions make it possible to write the cross section, for the process described by Eq. (1), as a product of the cross section for formation of the CC [Eq. (5)] multiplied by the probability that it decomposes into products. This leads to the following state-to-state cross section

$$\sigma(\bar{v}r^i, E_{trans}^i; \bar{v}r^f, E_{trans}^f) \\ = \frac{\pi \hbar^2}{2\mu_{A,B} E_{trans}^i \prod_k^{A,B} (2j^k + 1)} \sum_{i=0}^{l_{max}^i} \sum_{\left\{ \begin{array}{l} J \\ \bar{J} \text{ and } E_{tot} \text{ conserved} \end{array} \right\}} \\ \times \sum_{\left\{ \begin{array}{l} l^f \\ \bar{J} \text{ and } E_{tot} \text{ conserved} \end{array} \right\}} (2J + 1) P(J, E_{tot}; \bar{v}r^f, l^f, E_{trans}^f). \quad (6)$$

The conservation of energy (Assumption I) is ensured because the probability of decomposing the CC depends on E_{tot} .

Applying the principle of detailed balancing to Eq. (6), use the fact that $P(J, E_{tot}; \bar{v}r^f, l^f, E_{trans}^f)$ is independent of the initial states, and $P(J, E_{tot}; \bar{v}r^i, l^i, E_{trans}^i)$ is independent of the final states, and that the probability for the decomposition of a CC is proportional with the number of states that can be formed from a given channel (proved by Light in Ref. [6]). Light [6] has shown that the total cross section, Eq. (6), can be written:

$$\sigma(\bar{v}r^i, E_{trans}^i; \bar{v}r^f, E_{trans}^f) = \frac{\pi \hbar^2}{2\mu_{A,B} E_{trans}^i \prod_k^{A,B} (2j^k + 1)} \\ \times \sum_{J=0}^{\infty} (2J + 1) \frac{N(\bar{v}r^i, E_{tot}, J) N(\bar{v}r^f, E_{tot}, J)}{\sum_{x=\alpha, \beta, \dots} N(\bar{v}r^x, E_{tot}, J)}. \quad (7)$$

The reaction cross section given here depends on a specific chosen initial and final relative velocity between the molecules and their ro-vibrational state. This cross section can be summed in different ways to get less complete distributions. For example, as-

suming a Boltzmann distribution over the initial ro-vibrational states, and then summing up over the initial and final ro-vibrational quantum numbers, the total reaction cross section for the process is given by

$$\sigma(E_{tot}, T) = \frac{\pi \hbar^2}{2\mu_{A,B}} \frac{1}{\prod_k^{A,B} Q_{rot}^k Q_{vib}^k} \sum_{J=0}^{\infty} (2J+1) \times \sum_{\bar{v}r^i} \left(\frac{N(\bar{v}r^i, E_{tot}, J)}{E_{trans}^i} \right) \frac{\sum_{\bar{v}r^f} N(\bar{v}r^f, E_{tot}, J)}{\sum_{x=\alpha, \beta, \dots} \sum_{\bar{v}r^x} N(\bar{v}r^x, E_{tot}, J)}. \quad (8)$$

That means that in Eq. (8), the different $(\bar{v}r^i)$ cross sections are weighted with the rotational degeneracy of the two molecules divided by the rotational Q_{rot}^k and vibrational Q_{vib}^k partition functions of molecule A and B, respectively.

The thermal rate constant from a given initial to final ro-vibrational state is given by averaging the total state to state reaction cross sections with respect to the Boltzmann distribution

$$k(\bar{v}r^i; \bar{v}r^f; T) = S \sqrt{\frac{8k_b T}{\pi \mu_{A,B}}} \int_0^{\infty} d(E_{trans}^i / k_b T) \times \exp(-E_{trans}^i / k_b T) (E_{trans}^i / k_b T) \times \sigma(\bar{v}^i, \bar{j}^i, E_{trans}^i; \bar{v}^f, \bar{j}^f, E_{trans}^f) \quad (9)$$

where k_b is the Boltzmann constant, S is the symmetry factor for the reaction, and T is the temperature. The overall rate constant can be estimated by assuming a Boltzmann distribution over the initial ro-vibrational states, and then summing up over all the possible ro-vibrational quantum numbers. This gives

$$k(T) = S \sqrt{\frac{8k_b T}{\pi \mu_{A,B}}} \int_0^{\infty} d(E_{tot} / k_b T) \times \exp\left(\frac{-E_{tot}}{k_b T}\right) P(E_{tot}, T) \quad (10)$$

where

$$P(E_{tot}, T) = \frac{\pi \hbar^2}{2\mu_{A,B} k_b T} \frac{1}{\prod_k^{A,B} Q_{rot}^k Q_{vib}^k} \times \sum_{J=0}^{\infty} (2J+1) \frac{\sum_{\bar{v}r^i} N(\bar{v}r^i, E_{tot}, J) \sum_{\bar{v}r^f} N(\bar{v}r^f, E_{tot}, J)}{\sum_{x=\alpha, \beta, \dots} \sum_{\bar{v}r^x} N(\bar{v}r^x, E_{tot}, J)}. \quad (11)$$

Notice that Eqs. (7)–(11) only apply to collisions between two molecules. However, Eqs. (7) and (8) can also be used for atom-molecule reactions by removing all the ro-vibrational terms of the species that corresponds to the atom.

EVALUATION OF $N(E_{tot}, J)$

In this subsection we describe how the total number of states, $\sum_{\bar{v}r^x} N(\bar{v}r^x, E_{tot}, J) = N(E_{tot}, J)$, accessible from a CC can be calculated for different kinds of molecular systems.

We use the following notation: v_{max} = largest allowed vibration quantum number (see Table III); j_{max} = largest allowed rotational quantum number (see Table IV), and l_{max} = largest allowed orbital angular momentum quantum number (later we describe how this quantum number is assigned).

As stated in Assumption 1, some of the states of the system are prescribed, that is, E_{tot} and $\bar{J}$. In the following, for systems with two or three reaction partners, the equations for the total number of states, $N(E_{tot}, J)$, at a given total energy and total angular momentum, are evaluated. The chemical systems involve atoms, diatomic molecules, linear triatomic molecules, and molecules having a spherical, symmetric, or asymmetric top symmetry.

Atom + Molecule Collisions

Consider the case where A represents an atom and B a molecule composed of N atoms. The total energy of such a system is

$$E_{tot} = E_{trans}^{A,B} + E_{vib}^B + E_{rot}^B + V \quad (12)$$

and the total angular momentum is

$$\bar{J} = \bar{l}^{A,B} + \bar{j}^B. \quad (13)$$

The expressions for the vibrational and rotational energy for different types of molecules are given in Tables I and II. From Eq. (12), we then have the available energy

$$E_{avab} = E_{tot} - E_{vib}^B - V. \quad (14)$$

The first step in the evaluation of $N(E_{tot}, J)$ is to calculate the number of accessible states when the vibrational motion of the molecule is constant $N(\bar{v}^B, E_{tot}, J)$. From the equations for the quantum mechanical rotational energy in Table II, we see that if B is a diatomic, linear triatomic, or spherical

TABLE III

Expressions describing how to select the maximum vibrational quantum number v_{max} for various molecules taking part in a molecule collision. E_{tot} and V are the total energy and the potential energy for the system under investigation.

Diatomic Molecule

$$v_{max} = \frac{E_{tot}-V}{\hbar\omega} - \frac{1}{2},$$

where ω is the vibrational frequency for the diatomic molecule.

Linear Triatomic Molecule

$$v_{max_1} = \frac{E_{tot}-V}{\hbar\omega_1} - \frac{1}{2}$$

$$v_{max_{2a}}(v_1) = \frac{E_{tot}-E_{1a}-V}{\hbar\omega_2} - \frac{1}{2}$$

$$v_{max_{2b}}(v_1, v_{2a}) = \frac{E_{tot}-E_{1a}-E_{2a}-V}{\hbar\omega_2} - \frac{1}{2},$$

$$v_{max_3}(v_1, v_{2a}, v_{2b}) = \frac{E_{tot}-E_{1a}-E_{2a}-E_{2b}-V}{\hbar\omega_3} - \frac{1}{2},$$

where ω_i are the four vibrational frequencies and $i = 1, 2a, \text{ or } 2b$. E_{1a} , E_{2a} , and E_{2b} are defined in Table I.

Spherical, Symmetric, and Asymmetric Top Molecule

$$v_{max_1} = \frac{E_{tot}-V}{\hbar\omega_1} - \frac{1}{2}$$

$$v_{max_2} = \frac{E_{tot}-V-E_1}{\hbar\omega_2} - \frac{1}{2}$$

$\vdots$

$$v_{max_{s-1}} = \frac{E_{tot}-V-\sum_{i=1}^{s-2} E_i}{\hbar\omega_{s-1}} - \frac{1}{2}$$

$$v_{max_s} = \frac{E_{tot}-V-\sum_{i=1}^{s-1} E_i}{\hbar\omega_s} - \frac{1}{2},$$

where ω_i are the vibrational frequencies for the s oscillators. E_i is defined in Table I.

symmetric top molecule, the rotational energies are identical. Therefore, in this case, the number of accessible states is written as

$$N(\bar{v}^B, E_{avab}, J) = \sum_{j^B=0}^{j_{max}^B} \sum_{\left\{ \begin{array}{l} l^{A,B} = |J-j^B| \\ l^{A,B} \leq l_{max}^{A,B} \end{array} \right\}}^{J+j^B} 1 \quad (15)$$

where $l_{max}^{A,B}$ is given later in Eqs. (41) or (42). On the other hand, if B is a molecule with symmetric top symmetry, the expression of $N(\bar{v}^B, E_{avab}, J)$ becomes

$$N(\bar{v}^B, E_{avab}, J) = \sum_{j^B=0}^{j_{max}^B} \sum_{k^B=-j^B}^{j^B} \sum_{\left\{ \begin{array}{l} l^{A,B} = |J-j^B| \\ l^{A,B} \leq l_{max}^{A,B} \end{array} \right\}}^{J+j^B} 1. \quad (16)$$

TABLE IV

Expressions describing how to select the maximum rotational quantum number j_{max} and k_{max} for various molecules taking part in a molecule collision. E_a is the available energy for the vibrational degrees of freedom. The C and ε constants are defined in Table II. $\hbar$ is Planck's constant divided by 2π .

Diatomic, Linear Triatom, Spherical Top Molecule

$$j_{max}(j_{max}+1) = \frac{E_a}{C}$$

Symmetric Top Molecule

$$j_{max}(j_{max}+1) = \frac{E_a}{C_a}$$

$$(k_{max})^2 = \frac{E_a - C_a j(j+1)}{C_c - C_a}$$

Asymmetric Top Molecule

$j_{max}(j_{max}+1) = f(E_a)$ = the inverse of the expression for the energy of the asymmetric top molecule at $k=0$

$$k_{max}^2 = \frac{E_a - C_1 j(j+1) - C_2 j(j+1) \sum_{n=1}^{\infty} \varepsilon_{1n} C_3^n - C_2 j^2 (j+1)^2 \sum_{n=2}^{\infty} \varepsilon_{2n} (C_3)^n + \dots}{(C_2 + C_2 \sum_{n=2}^{\infty} \varepsilon_{0n} (C_3)^n)^{-1}}$$

When B is an asymmetric top molecule, it has three quantum numbers j , k , and k_c , where j and k correspond to the same quantum numbers as for the symmetric case. When $k = 0$, then $k_c = j$, otherwise k_c is equal to $j - k$ or $j - k + 1$. Let $\biguplus_{k_c^B}$ describe the summation over k_c as described above, then the number of accessible states when B is an asymmetric top molecule is given by

$$N(\bar{v}^B, E_{avab}, J) = \sum_{j^B=0}^{j_{max}^B} \sum_{k^B=-j^B}^{j^B} \biguplus_{k_c^B} \sum_{\substack{I+j^B \\ \{ \begin{smallmatrix} I^{A,B} = |J-j^B| \\ I^{A,B} \leq I_{max}^{A,B} \end{smallmatrix} \}}} 1. \quad (17)$$

The final step is to perform the summation over the vibrational motion of the molecules. If we define the following terms:

$$\begin{aligned} S_1^{diatom} &= \sum_{v_{diatom}=0}^{v_{max}^{diatom}} 1 \\ S_4^{triatom} &= \sum_{v_1^{triatom}=0}^{v_{max1}^{triatom}} \sum_{v_{2a}^{triatom}=0}^{v_{max2a}^{triatom}} \sum_{v_{2b}^{triatom}=0}^{v_{max2b}^{triatom}} \sum_{v_3^{triatom}=0}^{v_{max3}^{triatom}} 1 \\ S_s^{Sp} &= \left(\prod_{l=1}^s \sum_{v_l^{Sp}=0}^{v_{maxl}^{Sp}} \right) 1 \\ S_s^{Sy} &= \left(\prod_{l=1}^s \sum_{v_l^{Sy}=0}^{v_{maxl}^{Sy}} \right) 1 \\ S_s^{Asy} &= \left(\prod_{l=1}^s \sum_{v_l^{Asy}=0}^{v_{maxl}^{Asy}} \right) 1 \end{aligned} \quad (18)$$

we can write N for all the molecules in the following way:

$$N(E_{tot}, J) = S_t^B N(\bar{v}^B, E_{avab}, J) \quad (19)$$

where t is either 1, 4, or s , and B is either a diatomic, triatomic, spherical top (Sp), symmetric top (Sy), or asymmetric top (Asy).

Molecule + Molecule Collisions

The expression for the number of accessible states for the collisions between two molecules is more complicated. This arises from the fact that the total angular momentum depends on two rotational angular momenta—one from molecule A and one

from molecule B. The total angular momentum is

$$\bar{J} = \bar{I}^{A,B} + \bar{J}^A + \bar{J}^B. \quad (20)$$

For the atom-molecule case, we should find vectors of $\bar{I}^{A,B}$ and $\bar{J}^B$, where their sum was equal to $\bar{J}$. Since $\bar{J}$ consists of only two vectors, we could place them in the two dimensional space and then find the necessary vectors. In the case of a molecule-molecule collision, we must consider three vectors, and the vectors must be placed in the three dimensional space to calculate the total angular momentum. Handling such a problem is rather complicated and not very practicable. We can avoid this problem by simply rewriting Eq. (20) as

$$\bar{J} = \bar{I}^{A,B} + \bar{w} \quad \text{and} \quad \bar{w} = \bar{J}^A + \bar{J}^B. \quad (21)$$

Thus, if we find all possible vectors, $\bar{w}$, with the only restriction that $j^A \leq j_{max}^A$ and $j^B \leq j_{max}^B$, and then find the vectors that fulfill $\bar{J} = \bar{I}^{A,B} + \bar{w}$, we have circumvented the problem by reformulating the molecule-molecule case to be the same as the atom-molecule case. Therefore,

$$N(\bar{v}^A, \bar{v}^B, E_{avab}, J, w) = \sum_{j^A=0}^{j_{max}^A} \sum_{j^B=0}^{j_{max}^B(j^A)} 1 \quad (22)$$

if A and B are either diatomic, triatomic, or spherical top molecules. If one of the molecules (e.g., A) is a symmetric top, Eq. (22) becomes

$$N(\bar{v}^A, \bar{v}^B, E_{avab}, J, w) = \sum_{j^A=0}^{j_{max}^A} \sum_{k^A=-j^A}^{j^A} \sum_{j^B=0}^{j_{max}^B(j^A)} 1. \quad (23)$$

If both A and B are symmetrical top molecules, we have

$$N(\bar{v}^A, \bar{v}^B, E_{avab}, J, w) = \sum_{j^A=0}^{j_{max}^A} \sum_{k^A=-j^A}^{j^A} \sum_{j^B=0}^{j_{max}^B(j^A)} \sum_{k^B=-j^B}^{j^B} 1 \quad (24)$$

where w in Eqs. (22)–(24) is given by $\sqrt{j^{A2} + j^{B2} - 2j^A j^B \cos(\theta)}$,⁵ and θ is the angle between $\bar{I}^{A,B}$ and $\bar{w}$. Now adding the $\bar{I}^{A,B}$ vector to the equations and integrating over w , we can write

⁵We do not give the three expressions for the cases where B, or both A and B, can be asymmetric top molecules; but comparing Eqs. (22)–(24) with Eqs. (15)–(17) and (31)–(32), these expressions are easily obtained.

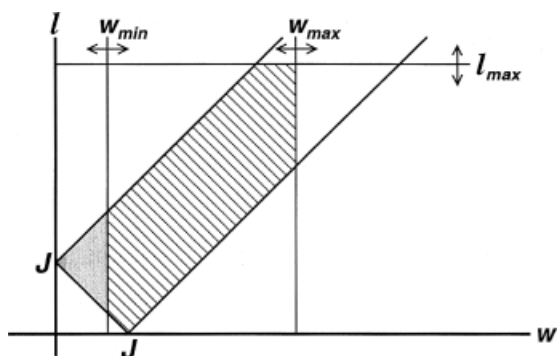


FIGURE 1. Region of integration in the l - w plane given by the double integral in Eq. (25), i.e., when the summation in Eq. (25) is replaced by an integral.

the total number of accessible states as

$$N(\bar{v}^A, \bar{v}^B, E_{avab}, J) = \int_{w_{\min}}^{w_{\max}} N(\bar{v}^A, \bar{v}^B, E_{avab}, J, w) \times \sum_{\substack{J+w \\ l^{A,B} \leq l_{\max}^{A,B}}} dw \quad (25)$$

where the last sum removes all the “forbidden” states. If the summation in Eq. (25) is replaced by an integral, the resulting double integral can easily be integrated analytically. The shaded and hatched area shown in Figure 1 illustrates the volume of integration.

Finally, we have to sum over the vibrational motion of the molecules. Because of the definitions given in Eq. (18), this will be

$$N(E_{tot}, J) = S_{t_1}^A S_{t_2}^B N(\bar{v}^A, \bar{v}^B, E_{avab}, J). \quad (26)$$

A + B + C Molecule Collisions

In the next section, it is discussed how the tri-molecular case (A + B + C) is handled. Here we will not derive the equations for all three molecule/atom cases; instead, we will present the equations for the case where the three molecules are treated as asymmetric tops. It is relatively easy to revise the equations for asymmetric tops to apply them to simpler systems with different symmetry point groups or to an atom. Following the concept used for the molecule-molecule case, we write the total angular momentum

$$\bar{J} = \bar{l}^{A,B} + \bar{l}^{AB,C} + \bar{j}^A + \bar{j}^B + \bar{j}^C \quad (27)$$

and the total energy as

$$E_{tot} = E_{trans}^{A,B} + E_{trans}^{AB,C} + E_{vib}^A + E_{vib}^B + E_{vib}^C + E_{rot}^A + E_{rot}^B + E_{rot}^C + V. \quad (28)$$

Making the same vector splitting as before, e.g.,

$$\begin{aligned} \bar{J} &= \bar{l}^{A,B} + \bar{w}_1, & \bar{w}_1 &= \bar{l}^{AB,C} + \bar{w}_2, \\ \bar{w}_2 &= \bar{j}^A + \bar{w}_3, & \text{and} & \quad \bar{w}_3 = \bar{j}^B + \bar{j}^C \end{aligned} \quad (29)$$

and we denote the angles between the vectors as

$$\begin{aligned} \theta_1 &= \angle(\bar{l}^{A,B}, \bar{w}_1), & \theta_2 &= \angle(\bar{l}^{AB,C}, \bar{w}_2), \\ \theta_3 &= \angle(\bar{j}^A, \bar{w}_3), & \text{and} & \quad \theta_4 = \angle(\bar{j}^B, \bar{j}^C) \end{aligned} \quad (30)$$

we obtain for the total number for accessible states

$$\begin{aligned} N(\bar{v}^A, \bar{v}^B, \bar{v}^C, E_{avab}, J) &= \int_0^{2\pi} d\theta_1 \int_0^{2\pi} d\theta_2 \int_0^{2\pi} d\theta_3 \int_0^{2\pi} d\theta_4 \\ &\times \sum_{j^A=0}^{j_{\max}^A} \sum_{k^A=-j^A}^{j^A} \bigoplus_{k_C^A}^{j_{\max}^B(j^A)} \sum_{j^B=0}^{j^B} \sum_{k^B=-j^B}^{j^B} \bigoplus_{k_C^B}^{j_{\max}^C(j^A, j^B)} \sum_{j^C=0}^{j^C} \bigoplus_{k_C^C}^{j_{\max}^{AB,C}} \sum_{l^{AB,C}=0}^{J+w_1} \sum_{\substack{l^{A,B}=|J-w_1| \\ l^{A,B} \leq l_{\max}^{A,B}}} 1. \end{aligned} \quad (31)$$

If the vibrational motion of the molecules is also included, we get

$$N(E_{tot}, J) = S_{t_1}^A S_{t_2}^B S_{t_3}^C N(\bar{v}^A, \bar{v}^B, \bar{v}^C, E_{avab}, J). \quad (32)$$

Using this methodology, it is simple to extend the equations to more advanced systems.

Assignment of Maximum Orbital Angular Momentum

The most critical part of the theory lies in the choice of the requirements for a CC. As stated in the previous section, a CC is formed when the kinetic energy of the A–B system can exceed the barrier of the effective potential, and the orbital angular momentum is less than or equal to the maximum orbital angular momentum. In this section, two kinds of chemical reactions will be discussed—reactions with and without potential energy barriers. The methodology to calculate the maximum orbital angular momentum is taken from the work by Light and coworkers [1–6].

REACTIONS WITHOUT ENERGY BARRIERS

Reactions without energy barriers are the simplest type to handle. The impact parameter, b and l are related through the relationship $b^2 = l(l+1)\hbar^2/2\mu E_{trans}$. For reactions without energy barriers, there exists a distance r^* where the molecule/atom-molecule is strongly coupled. If the reaction has no energy barrier, it is reasonable to believe that r^* is approximately independent of the two compounds' internal energy, relative translational energy between them, and their rotational angular momentum. If it is assumed that the maximum of the effective potential occurs in that attractive region of the potential energy surface, the short-range part of the potential can be neglected. Perturbation studies have shown that the intermolecular forces describing the long-range potential follows the expression

$$V_{LR} \approx -CR^{-n} \quad (33)$$

where $n = 4$ for ion-molecule reactions and $n = 6$ for neutral reactions. The magnitude of C , for neutral molecules, is the spherically averaged sum of dipole-dipole, dipole-induced-dipole, and dispersion contributions. Therefore, the effective potential between the molecules in the channels is

$$V_{eff} = V_{LR} + \frac{l(l+1)\hbar^2}{2\mu}R^{-2}. \quad (34)$$

From this potential, we can evaluate at which R the effective potential has its maximum value. The maximum angular momentum, $l_{max}(l_{max}+1)\hbar^2$, is then fixed by

$$l_{max}(l_{max}+1)\hbar^2 = n\mu C^{2/n}(2E_{trans}/(n-2))^{1-2/n}. \quad (35)$$

This equation is most applicable to low energy conditions; but at higher energies, the maximum impact parameter, b_{max} , given by this equation can be lower than r^* . This should not be allowed. Therefore, it is more reasonable to let b_{max} be the maximum of these two values. The result, expressed in maximum orbital angular momentum [2-4], is

$$l_{max}(l_{max}+1)\hbar^2 = \max \left\{ \begin{array}{l} n\mu C^{2/n}(2E_{trans}/(n-2))^{1-2/n} \\ 2\mu E_{trans}(nC - 2C)^{2/n}(2E_{trans})^{2/n} \end{array} \right. \quad (36)$$

Evaluation of the Long-Range Potential

C [Eq. (36)] for neutral molecules is the spherically averaged sum of dipole-dipole, dipole-

induced-dipole, and dispersion intermolecular interactions.

The thermally averaged dipole-dipole interaction is given by [29]:

$$W_{dd}(r) = -\frac{2}{3} \frac{u_A^2 u_B^2}{k_b T (4\pi \epsilon_0)^2} r^{-6} \quad (37)$$

where ϵ_0 is the permittivity of vacuum and r denotes the distance between the two dipoles. The two molecules A and B have a permanent dipole moment, and they are denoted u_A and u_B , respectively.

The next type of interaction arises from the polarization or induced dipole moment on molecule B (independent of whether B is polar or nonpolar) due to the field from the dipole moment of molecule A. The thermally averaged interaction is given by [29]

$$W_{di}(r) = -\frac{u_A^2 \alpha_B}{(4\pi \epsilon_0)^2} r^{-6}. \quad (38)$$

If both molecules are polar, the equation becomes [29]

$$W_{di}(r) = -\frac{u_A^2 \alpha_B + u_A^2 \alpha_A}{(4\pi \epsilon_0)^2} r^{-6} \quad (39)$$

where α_B is the isotropic polarizability of molecule A or B.

Finally, we consider the interactions between two nonpolar units (the London dispersion forces). The dispersion interactions relate to the fluctuations in the electronic distributions of the molecules, and its evaluation requires a quantum mechanical treatment in which the interactions are taken into account by second order perturbation theory. London developed the following approximate formula for the dispersion interactions [29]:

$$W_{ii}(r) = -\frac{3}{2} \frac{\alpha_A \alpha_B}{(4\pi \epsilon_0)^2} \frac{E_A^I E_B^I}{E_A^I + E_B^I} r^{-6} \quad (40)$$

where E_x^I is the ionization energy of molecule A or B. The ionization energy can be calculated using Koopman's theorem, where the ionization energy is given as the negative of the highest occupied orbital energy.

If both molecules are polar, the long-range potential is a summation of these three intermolecular interactions.

REACTIONS WITH ENERGY BARRIERS

It is more difficult to make an unambiguous theory for reactions with energy barriers. In this case, it is reasonable to assume that the formation of a CC depends on the energy barrier, E_b , r^* [which

is the distance between the center of mass of the reacting molecules at $E_b(r^*)$, the translational energy, the molecules' ro-vibrational energy, etc. For the systems we are interested in, it is not possible to calculate directly this information because this would demand more detailed PES information. The model is therefore developed from a reasonable guess [5, 6]. We set

$$l_{\max}(l_{\max} + 1)\hbar^2 = \begin{cases} (E_{\text{avab}} - E_b - \frac{j_B(j_B+1)\hbar^2}{2I_B} - \frac{j_A(j_A+1)\hbar^2}{2I_A})2\mu_{A,B}r^{*2} & \text{if } E_b \geq 0 \\ (E_{\text{avab}} - \frac{j_B(j_B+1)\hbar^2}{2I_B} - \frac{j_A(j_A+1)\hbar^2}{2I_A})2\mu_{A,B}r^{*2} & \text{if } E_b < 0 \end{cases} \quad (41)$$

where $E_{\text{avab}} = E_{\text{tot}} - E_{\text{vib}} - V$.

$l_{\max}$ FOR N-MOLECULAR/ATOMIC SYSTEMS

From a theoretical point of view, a reaction can always be split into a number of bimolecular elementary reactions; therefore, a theoretical description of atom-molecule and molecule-molecule reactions should be a complete description of all chemical reactions. The systems we are interested in are molecular collisions of LMS. Experiments can be used to estimate reactants and products for some of these systems, but finding the reaction mechanism and the transition state is very difficult, e.g., the self-reaction of the simplest peroxy radical. This reaction has been investigated by Lightfoot et al. [30, 31]. They could not estimate the reaction pathway, but they were able to predict the products. This creates some problems because even though the reaction starts with the collision of two molecules, the products result from the composition of three molecules, and the mechanism by which the molecules interact to form these products is unknown. Therefore, we also have to discuss which kind of concept can be used to estimate $l_{\max}$ for N-molecule/atom systems. We will not discuss the case for the general N-molecule/atom case, even though it is possible, because it will be much more illustrative to present the model for the three-molecule/atom case ($A + B + C$). From this case, it is simple to expand the model to more advanced systems.

For an $A + B + C$ system, we can have the following orbital angular momenta: $(l^{A,B}, l^{C,AB})$, $(l^{A,C}, l^{B,AC})$, or $(l^{B,C}, l^{A,BC})$. Let us assume that we have sufficient information to calculate $l_{\max}$ for each of these orbital

angular momenta, then we define

$$l_{\max}(l_{\max} + 1) = \max \begin{cases} \min\{l_{\max}^{A,B}(l_{\max}^{A,B} + 1), l_{\max}^{C,AB}(l_{\max}^{C,AB} + 1)\} \\ \min\{l_{\max}^{A,C}(l_{\max}^{A,C} + 1), l_{\max}^{B,AC}(l_{\max}^{B,AC} + 1)\} \\ \min\{l_{\max}^{B,C}(l_{\max}^{B,C} + 1), l_{\max}^{A,BC}(l_{\max}^{A,BC} + 1)\} \end{cases} \quad (42)$$

The orbital angular momenta of the channel bottle-necks are found by taking

$$\min\{l_{\max}^{N_1,N_2,N_3}(l_{\max}^{N_1,N_2,N_3} + 1), l_{\max}^{N_3,N_1,N_2}(l_{\max}^{N_3,N_1,N_2} + 1)\}$$

for each channel. The reaction bottleneck is defined as the maximum value of these channel bottlenecks, $l_{\max}$; therefore, the reaction bottleneck has the highest reaction probability to form a CC.

By making such a definition of $l_{\max}$, the expression of $l_{\max}$ is the same as we use for bimolecular reactions.

Discussion

In this article, a new theoretical model based upon phase-space theory has been presented. An advantage of this model is that only a very limited number of input parameters are needed to calculate the rate constants for a reaction if the reactant/product channels are known. Spectroscopic data for the molecules existing in the reactant/product channels and their potential energy are required to calculate the number of states. To calculate the maximum impact parameter, either the energy barrier is required or—if the reaction channels do not have a barrier—the dipole moments, polarization, and ionization energy for the molecules are required.

The presentation of phase-space theory given in this article is based upon a discrete expression of the theory, while those given in Refs. [1–6] and [19–24] present a continuous description. We believe that the increased capability of present computer technology allows the use of a discrete and less approximate representation [25], and a continuous description of the interacting species vibrational energy if the reaction does not have an energy barrier is physically wrong [25].

If it is possible to use full reaction dynamics, reaction path dynamics, or transition state theory to calculate the rate constant for a molecular system, then these theoretical methods will, of course, be preferable. Unfortunately, for these methods, the PES, the reaction path, or information on the transition state are available for a very limited number

of molecular systems, and usually these are systems composed only of light elements.

The model requires either measured or calculated spectroscopic data for stable molecules/atoms. Although the required spectroscopic data have been measured for only a very limited number of the molecules of interest to atmospheric chemistry, combustion, and pyrolysis, the model may use the results of molecular electronic structure calculations with good accuracy. We predict that this approximate approach has very little effect on the calculated total reaction cross section and rate constant, because the molecular properties are incorporated in both the numerator and denominator of the total reaction cross section [Eqs. (7) and (8)].

The greatest approximation in our phase-space model concerns the formation and decomposition of the CC into the reaction channels. Even though Assumptions II and III appear to be logical from a classical point of view, classical dynamical simulations show [32, 33] that not every trajectory in compliance with Assumption II actually leads to the formation of a CC. Therefore, this method will tend to overestimate the probability of forming a CC, unless there are competing quantum effects such as tunneling. Furthermore, many full dynamical simulations indicate that for triatomic systems most reactive collisions are almost colinear (see, e.g., Ref. [33]); e.g., Assumption III also overestimates the reaction probability. These considerations have not been made in our model because we wanted the model to require the fewest potential energy points or data concerning molecular properties as possible. Consideration of the two problems described requires further information about potential energy surfaces or molecular properties. For triatomic systems, it is possible to assign a geometric parameter that only allows almost colinear collisions to be reactive, but making a conservative assignment of such a parameter for LMS is not possible for the general case.

The method does not require the localization of the transition state configuration which is not easy to localize for a complex molecular system with a large number of intramolecular degrees of freedom. The application of the transition state expression of the rate constant requires this knowledge. Traditionally, there is no automatic procedure for obtaining transition state configurations for bimolecular reactions, and it is often observed that when calculating the Hessian for a potential transition state configuration, one obtains more than one negative eigenvalue of the diagonal Hessian matrix.

In future investigations, we focus on reactions that have negative activation energies where the use of transition state theory for calculating the rate constants is not possible. Phase-space methods are able to treat this class of chemical reactions and are expected to do well. This has been observed in the work of Light and co-workers.

As well as predicting the distribution of product states from thermal reactions, ionic fragmentation, and photochemical unimolecular dissociation, the phase-space theory can be used to calculate state-to-state cross sections [Eq. (7)] and rate constants [Eq. (9)], and total cross sections [Eq. (8)] and rate constants [Eq. (10)] for bimolecular reactions proceeding through long-lived complexes. In Ref. [25], the presented model has been tested for HO radical atmospheric reaction, and a further description of the usefulness of the model is given.

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